

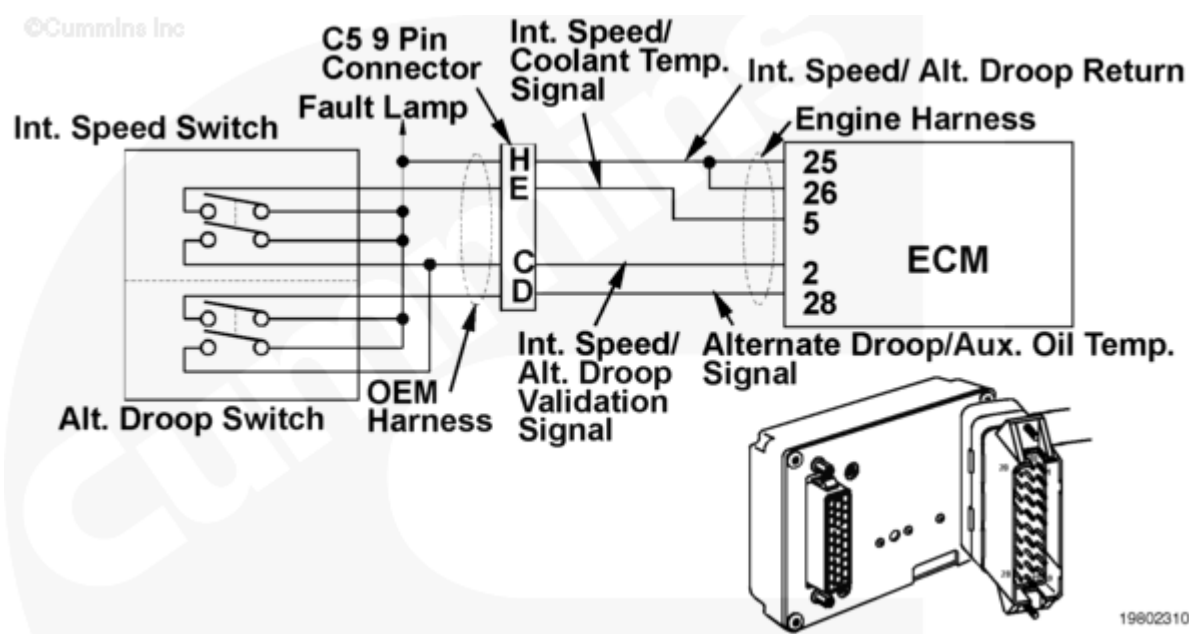
Fault Code: 523

Auxiliary Intermediate (PTO) Speed Switch Validation - Data Erratic, Intermittent, or Incorrect

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 523 PID(P): Pa SPN: FMI: 2 Lamp: On SRT: 00-647	The intermediate-speed/alternate droop validation signal is off when the intermediate-speed signal is on.	Loss of intermediate-speed, or engine will run at intermediate-speed only .



Intermediate-Speed/Alternate Droop Validation Signal Circuit

Circuit Description

The intermediate-speed/alternate droop validation signal is a safety feature that disables the intermediate-speed feature while the validation signal is detected as a high voltage. The intermediate-speed switch provides the intermediate-speed and intermediate-speed/alternate droop validation signals to the ECM through the OEM harness and the main engine harness.

Component Location

The intermediate-speed switch and the alternate droop switch location may vary and is OEM dependent.

Conditions For Running The Diagnostics

This diagnostic runs continuously when the engine is running.

Conditions For Setting The Fault Codes

The Engine Control Module (ECM) detected high voltage at the intermediate-speed/alternate droop validation signal when low voltage is detected at the intermediate-speed signal.

Action Taken When The Fault Code Is Active

- The ECM illuminates the amber CHECK ENGINE light when the diagnostic runs and fails.
- No engine protection for engine oil pressure.

For Power Generation Applications:

- The generator set controller displays the fault immediately when the diagnostics runs and fails.
- No engine protection for engine engine oil pressure.

Conditions For Clearing The Fault Code

- To validate the repair, perform a key cycle, start the engine and let it idle for 1 minute.
- The fault code status displayed by the recommended Cummins® electronic service tool or equivalent will change to INACTIVE immediately after the diagnostic runs and passes.
- The ECM will turn off the amber CHECK ENGINE lamp immediately after the diagnostic runs and passes.
- The “Reset All Faults” command in the recommended Cummins® electronic service tool or equivalent can be used to clear active and inactive faults.

For Power Generation Applications:

- To validate the repair, start the engine and let it run for 1 minute at no load.
- The generator set controller will turn off the indicator immediately after the user presses reset.
- The “Reset All Faults” command in the recommended Cummins® electronic service tool or equivalent can be used to clear active faults.

Shoptalk

Possible causes of this fault code include:

- Malfunctioning or damaged engine wiring harness.
- Malfunctioning or damaged OEM wiring harness.
- Damaged or loose connectors.
- Malfunctioning or damaged intermediate-speed switch (retard).
- Malfunctioning or damaged alternate droop switch.

Refer to Troubleshooting Fault Code t05-523 (</qs3/pubsys2/xml/en/procedures/98/98-t05-523.html>)

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